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Issac Sim

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Issac Sim

Overview

NVIDIA Issac Sim is a robotics simulation platform built on Omniverse for photorealistic rendering, GPU-accelerated physics, and high-fidelity sensor simulation NVIDIA, 2022. It supports reinforcement learning, motion planning, synthetic data creation, and full-stack robotics workflows with Jetson-based robots.

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Hardware Requirements for NVIDIA Issac Sim I

NVIDIA Issac Sim is a GPU-accelerated robotics simulator built on USD, RTX ray tracing, and PhysX 5. Therefore, hardware must support real-time ray tracing, high-fidelity physics, and large-scale scene rendering Engineering, 2024.

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Minimum Requirements (Basic Scenes) I

- **GPU:** NVIDIA RTX 3060 or Quadro RTX 3000
- **CUDA Compute Capability:** 7.5+
- **VRAM:** 6–8 GB
- **CPU:** Quad-core Intel i5 / AMD Ryzen 5
- **RAM:** 16 GB
- **Storage:** 50 GB SSD
- **OS:** Ubuntu 20.04 / 22.04 or Windows 10/11

Recommended Requirements (Robotics, SLAM, Multi-Robot Scenes) I

- **GPU:** NVIDIA RTX 3080, RTX 4080, or L40 (industrial)
- **VRAM:** 12–16 GB for heavy textures + synthetic data
- **CPU:** 8-core Ryzen 7 / Intel i7 or higher
- **RAM:** 32 GB or more
- **Storage:** 200 GB NVMe SSD for USD assets + Omniverse cache
- **Network:** 1 Gbps for Omniverse Nucleus collaboration

High-End Requirements (Photorealistic Ray-Tracing, Digital Twin Workflows) I

- **GPU:**
  - NVIDIA RTX 4090 for desktop
  - NVIDIA A6000/A8000 for workstation
  - NVIDIA H100 for datacenter simulation workloads
- **VRAM:** 24–48 GB
- **CPU:** 16-core Threadripper / Xeon workstation
- **RAM:** 64–128 GB for multi-robot + large environment

RAM and Computation Limitations I

- NVIDIA Issac Sim is heavily optimized for **RTX ray tracing cores**.
- Scenes involving sensors (RGB, depth, IMU, segmentation) significantly increase VRAM usage.
- Large USD worlds require high disk throughput (fast NVMe recommended).
- Running Issac Sim on laptops is supported, but GPU thermals may limit performance.

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Installation Steps for NVIDIA Issac Sim I

1. Install NVIDIA Omniverse Launcher Team, 2023.
2. Download NVIDIA Issac Sim from the Launcher.
3. Install dependencies:
   - Python 3.8/3.10
   - NVIDIA GPU driver + CUDA
4. Verify installation using:

```
./isaac-sim.sh
```

Packages

NVIDIA Issac Sim Packages

- **Isaac Core API** — simulation control, physics management Robotics, 2023.
- **Isaac ROS** — bridges ROS/ROS2 with simulated sensors.
- **Isaac Replicator** — synthetic data and domain randomization.
- **Isaac Lab** — RL training environments.
- **Omnigraph** — node-based control pipelines.

Configuration of NVIDIA Issac Sim

Configuring NVIDIA Issac Sim correctly is essential for accurate robotics simulation, real-time performance, and sensor fidelity Team, 2024b. The configuration workflow includes:

- Application Settings
- Physics Engine Configuration
- Sensor Settings
- ROS2 Bridge Configuration
- Scene Setup and Environment Parameters
- Performance Optimization

Application Settings

1. **Render Settings:**
   - RTX Path Tracing or RTX Real-Time
   - Anti-aliasing: TAA or FXAA
   - Render Resolution (e.g., 1080p for real-time)
   - Shadow Quality and Material Sampling
2. **Simulation Settings:**
   - Time step: 60 Hz (default) or 120 Hz for high-speed robots
   - Substeps: 4–8 for stable wheel-ground contact
   - Gravity: -9.81 m/s^2
3. **Resource Management:**
   - Enable USD Asset Caching
   - Limit GPU texture memory if required

Physics Configuration (PhysX 5)

PhysX Settings

- Solver Type: Temporal Gauss-Seidel (TGS)
- Use GPU Dynamics: Recommended for large scenes
- Restitution, friction, damping settings adjusted per robot

Wheel Robot Configuration:

- Enable articulations for wheel joints
- Use differential drive or custom motor model
- Set wheel radius + wheelbase for correct odometry

Ground Plane Configuration:

- High-friction surfaces for stable wheel traction
- Collision Mesh Resolution: Medium/High

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Sensor Configuration

Camera Configuration:

- Resolution: 640x480 or 1280x720
- Frame rate: 30–60 FPS
- Intrinsic parameters: fx, fy, cx, cy
- Extrinsic alignment to robot frame

IMU Configuration (if used):

- Gyroscope + accelerometer bias
- Noise models (Gaussian)

Depth or Segmentation Sensors:

- Enable depth clipping
- Adjust max distance (5–20 m)

ROS2 Bridge Configuration

The Isaac ROS2 bridge allows sensor and control topics to flow between the robot system and NVIDIA Issac Sim Developers, 2024.

Modules to Enable:

- `/rgb`, `/camera_info`
- `/cmd_vel` (Twist)
- `/odom` (Odometry)
- `/tf` and `/tf_static`

Example ROS2 Launch:

```
ros2 launch isaac_ros2_bridge isaac_sim_bridge.launch.py
```

Simulation Time Sync:

- Use `use_sim_time:=true`
- Enable ROS clock publishing

Environment and Scene Configuration

Scene Composition:

- Load USD environment (warehouse, room, outdoor)
- Insert obstacles for SLAM feature richness
- Adjust lighting for photorealism or performance

Robot Setup:

- Import URDF or USD robot model
- Configure joints, limits, inertia
- Attach sensors to correct robot frames

Navigation Environment:

- Add ArUco markers (optional)
- Ensure textured surfaces for VSLAM

Performance Optimization

Real-Time Simulation Tips:

- Lower render resolution (720p for speed)
- Reduce light bounces in path tracing
- Use RTX Real-Time instead of Path Tracing
- Disable unnecessary physics bodies

Memory Optimization:

- Reduce texture resolution of non-important assets
- Use instancing when duplicating objects

GPU Optimization:

- Enable DLSS (if available)
- Limit simulation to 60 FPS max

Simulated Environment Images

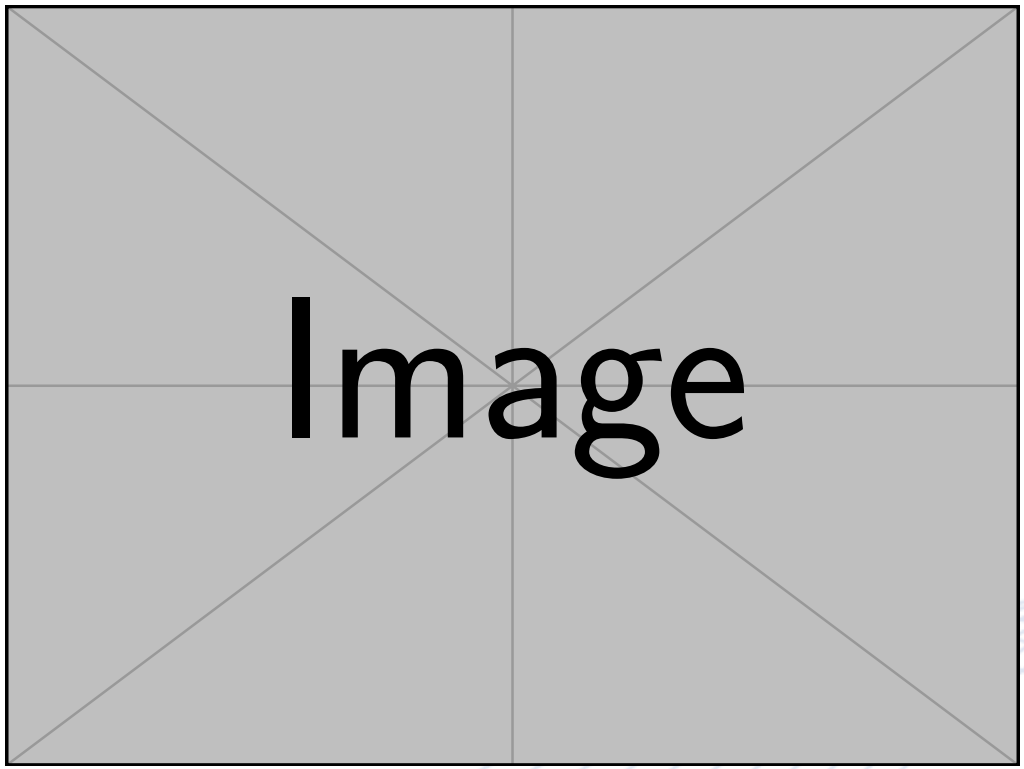


Abbildung: Example robot scene rendered in NVIDIA Issac Sim Lab, 2024.

Example schematic of robot coordinate frame in simulation Team, 2024a.

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Localization Slides

Robot Hardware Used

- NVIDIA Jetson Nano 4GB A02 NVIDIA, 2020.
- Raspberry Pi Camera V2 8MP Foundation, 2018.
- 2-Wheel 48:1 Dagu robot chassis + metal caster Robotics, 2019.
- ROS2 Humble or Foxy for communication.

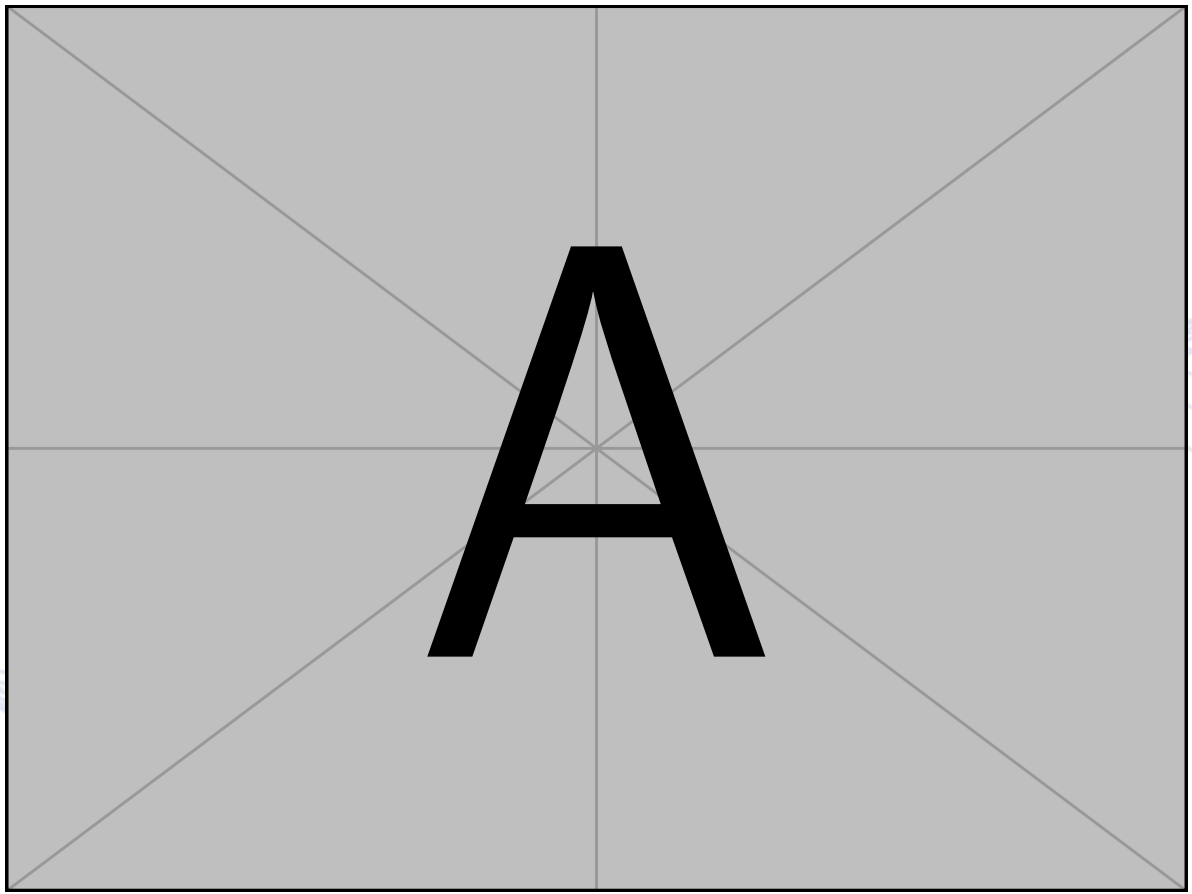
Why No LiDAR?

Localization is performed using only camera-based visual odometry and wheel-encoder fusion instead of LiDAR SLAM Robotics, 2022.
Advantages:

- Lower cost hardware
- Compatible with Jetson Nano
- Works well with Issac Sim synthetic camera feeds

Camera-Based Localization Pipeline

1. Raspberry Pi V2 Camera simulated in NVIDIA Issac Sim.
2. Visual Odometry nodes (Isaac ROS VSLAM).
3. Wheel odometry from motor encoders.
4. Fusion via Extended Kalman Filter (EKF).



Simulation Setup in NVIDIA Issac Sim I

Robot Modeling:

- Import URDF of 2-Wheel Dagu robot.
- Add camera prim at correct height + orientation.
- Configure differential drive controller.

Sensor Setup:

- Enable RGB camera + depth (if needed).
- Configure ROS2 bridge topics.

Environment:

- Indoor warehouse world.
- Add textures + obstacles for SLAM features.

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Simulation Setup in NVIDIA Issac Sim II

Testing Localization:

```
ros2 launch isaac_ros_visual_slam isaac_ros_visual_slam.launch.py rectified:=true use_sim_time:=true
```

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Vielen Dank

für Ihre Aufmerksamkeit

Quellen

References I

- Developers, NVIDIA Isaac ROS (2024). *ROS2 Bridge Setup for NVIDIA Issac Sim*.
- Engineering, NVIDIA Systems (2024). *NVIDIA Issac Sim Hardware and Performance Requirements*.
- Foundation, Raspberry Pi (2018). *Camera Module V2 Datasheet*.
- Lab, Simulation (2024). *Rendering Examples in NVIDIA Issac Sim*.
- NVIDIA (2020). *Jetson Nano Developer Kit A02 Specifications*.
- — (2022). *NVIDIA Issac Sim Overview*.
- Robotics, Dagu (2019). *Dagu 2-Wheel Differential Chassis*.
- Robotics, NVIDIA (2023). *Isaac Core API Reference*.
- Robotics, Open (2022). *Camera-Only Visual Odometry Techniques*.
- ROS, NVIDIA Isaac (2023). *Visual SLAM Pipeline Documentation*.
- Team, Graphics (2024a). *Coordinate Frame TikZ Diagram*.

References II

 Team, NVIDIA Omniverse (2023). *Omniverse Launcher Documentation*.

 Team, NVIDIA Robotics Simulation (2024b). *Configuration Guide for NVIDIA Issac Sim*.